When does graded shaping matter?

Cell = sampled success gap (graded − sparse), mean over three training seeds

Episode step budget

	4n	2n	1.25n
Small fixed n=8–16	+0.4 pp graded 99.7% · sparse 99.3% seed ranges 99.5–99.9 / 99.1–99.4	+1.2 pp graded 97.7% · sparse 96.5% seed ranges 97.3–98.0 / 96.0–97.1	+1.9 pp graded 95.1% · sparse 93.1% seed ranges 94.7–95.3 / 92.6–93.6
Cold start n=32–64	+7.7 pp graded 97.1% · sparse 89.3% seed ranges 96.3–97.6 / 86.7–90.9	+8.5 pp graded 89.1% · sparse 80.6% seed ranges 86.3–91.4 / 78.1–83.5	+11.5 pp graded 83.2% · sparse 71.7% seed ranges 82.1–85.2 / 70.3–74.3

HONEST NULL — no cell reaches ≥15 pp with non-overlapping seed ranges

Shared 1,000-graph suites per size regime; 5M-step LSTM/Muon policies; goal bonus = 10; no curriculum.